

Decoding Cognitive Performance from EEG Using Energy-Based and Biophysical Models

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Abstract

Understanding the neural mechanisms underlying cognitive performance requires models that go beyond descriptive EEG features toward interpretable representations of brain dynamics. In this study, we applied energy landscape analysis, attractor modeling, and neural mass simulations to EEG data from 36 individuals during rest and a paced counting task, with the goal of identifying neural markers distinguishing high and low performers.

Our energy-based analysis of alpha and beta amplitude envelopes suggested that good performers may exhibit broader energy ranges and more local minima, indicating a trend toward increased metastability. However, none of these comparisons reached statistical significance, and all high-density intervals (HDIs) included zero. Hopfield network-based attractor modeling yielded small, non-significant differences favoring the good group in attractor count and basin entropy across several bands, suggestive of richer but unconfirmed dynamical capacity.

In contrast, circuit-level modeling with the Jansen-Rit neural mass model revealed credible group differences in two parameters—recurrent connectivity strength (C) and background input (μ) both showing HDIs excluding zero. These effects were consistent across rest and task conditions and may reflect enhanced cortical excitability and coupling in high performing individuals, though multiple testing remains a caveat.

Finally, we trained an XGBoost classifier on extracted model features using synthetic balancing (SMOTE) to address class imbalance. The model achieved 81% test accuracy (95% CI: [62%, 94%]), indicating some predictive utility. However, this result lacks comparison to simpler baselines and should be interpreted as preliminary given the small and imbalanced dataset.

In sum, while most findings were exploratory and limited by sample size, certain converging trends point to the potential of integrative modeling for understanding individual differences in brain function. Future work should prioritize replication, statistical correction, and comparisons across simpler and more scalable methods to validate and refine these candidate neural markers.

1 Introduction

Electroencephalography (EEG) provides a temporally precise, non-invasive window into large-scale brain activity and has long been central to cognitive and clinical neuroscience. Despite its utility, EEG signals reflect coarse, low-dimensional projections of the brain’s high-dimensional dynamics,

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limiting the capacity of traditional analyses such as spectral power or connectivity to reveal mechanistic underpinnings of cognitive function [32, 14]. These conventional methods often capture statistical associations but offer limited insight into the latent dynamical processes that support perception, attention, or memory.

To address this gap, recent research has embraced modeling frameworks from dynamical systems theory and statistical physics to probe the structure and function of neural state spaces. One prominent approach is energy landscape analysis inspired by the Ising model, which treats brain activity patterns as states in a probabilistic system evolving over a rugged energy surface [16, 39]. This method allows researchers to identify metastable states, transition probabilities, and attractor basin configurations that may characterize different cognitive phenotypes. By applying Ising-based analysis to EEG amplitude envelope states, it is possible to quantify energy range, local minima count, and entropy—measures that reflect neural complexity and flexibility.

In parallel, attractor-based recurrent neural networks such as Hopfield models have been adapted for neuroscience applications beyond their original associative memory formulation [21, 1]. When applied to EEG time series, these models can reconstruct the underlying dynamical landscape by identifying stable patterns (attractors) and the convergence paths toward them [40]. Metrics such as attractor count, basin entropy, and convergence depth help to quantify how readily the brain transitions between neural states, a hallmark of cognitive adaptability.

To further link these network-level dynamics with physiological substrates, neural mass models like the Jansen–Rit framework offer biophysically grounded simulations of cortical activity [22]. This model captures the interactions between excitatory pyramidal neurons and inhibitory interneurons, and when fitted to subject-specific EEG data particularly in the alpha band it yields interpretable parameters such as synaptic gain and background input [41, 11]. These parameters serve as proxies for cortical excitability, E–I balance, and responsiveness, thereby offering mechanistic insights into individual variability in brain function.

These three modeling approaches were purposefully combined because they span complementary levels of abstraction. The Ising model offers a statistical physics perspective on macro-level brain state transitions and stability; the Hopfield model bridges this with a dynamical systems view of attractor structure in specific frequency bands; and the Jansen–Rit model anchors these phenomena in biologically interpretable parameters of cortical microcircuits. Together, they allow for a multiscale dissection of EEG dynamics moving from global energy topology to mesoscopic attractor dynamics and finally to biophysical circuit properties. This integrated approach provides a more complete account of how brain dynamics encode behavior than any single model in isolation.

In this study, we apply this hybrid modeling pipeline to EEG data collected from individuals during resting state and a sustained counting task. Participants were behaviorally stratified into high and low-performing groups based on task accuracy, yielding a natural performance contrast. From each modeling framework, we extracted neural features hypothesized to relate to cognitive performance: energy range and transition entropy from the Ising model; attractor count and basin size from the Hopfield model; and synaptic coupling and background input from the Jansen–Rit model. We then employed machine learning to assess whether these features could reliably classify individuals as good or poor performers, using cross-validation and class balancing to ensure robustness.

By combining data-driven modeling with mechanistic interpretation, our approach transcends surface-level EEG features to uncover biologically and dynamically meaningful correlates of cognitive performance. This framework contributes to the growing effort to translate neurophysiological signals into interpretable markers of brain function, with potential relevance for individualized neuroscience and clinical applications in cognitive assessment.

2 Methods

This study investigates inter-individual variability in cognitive performance by analyzing electrophysiological recordings acquired during an eyes-closed resting-state condition and a controlled mental arithmetic task. Data were collected from thirty-six adult participants using a 21-channel setup, which included 19 scalp EEG electrodes and 1 ECG channel, along with a bipolar reference (A2–A1), conforming to the international 10–20 system. Recordings were sampled at 500 Hz and covered frontal, central, temporal, parietal, and occipital regions. During the task condition, participants performed a serial subtraction operation under time pressure, and their behavioral scores were computed as the number of correctly subtracted responses within the allotted time. Based on performance, participants were classified into two groups: high performers (“good counters”, $n = 26$) and low performers (“bad counters”, $n = 10$), independent of their EEG data. EEG data collection and preprocessing were performed by Goldberger et al. using standardized protocols from PhysioNet’s open-access resources, which include validated tools for physiologic signal acquisition and analysis.[18]

The overarching goal of this work is to uncover neurophysiological markers of cognitive ability by modeling the underlying dynamics of EEG signals across conditions and frequency bands. We employ three computational modeling frameworks Hopfield attractor networks, Ising-based energy landscape analysis, and the Jansen-Rit neural mass model—each offering complementary perspectives on brain dynamics. These models were applied to bandpass-filtered EEG signals in standard frequency ranges: theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), gamma (30–45 Hz), and broadband (1–45 Hz). Each model generated subject- and condition-specific feature sets capturing aspects of network stability, attractor structure, and physiological dynamics.

Subsequently, we conducted group-level statistical comparisons to evaluate whether these features differentiated between good and bad performers. Finally, all extracted features were integrated into a unified classification framework to assess their predictive utility in identifying cognitive performance profiles. The following subsections describe in detail the data preprocessing, modeling strategies, feature extraction procedures, and statistical methodologies employed in this study.

2.1 Ising-Inspired Energy Landscape Analysis of EEG Brain States

To probe the dynamical structure of large-scale brain activity, we employed a statistical physics-based framework that models EEG-derived brain states as evolving within an abstract energy landscape. Drawing on the Ising model and pairwise maximum entropy principles, this approach estimates effective interactions between EEG channels and quantifies stability, complexity, and attractor structure of cortical dynamics [36, 31]. The method was applied to identify group-level differences between individuals classified as high and low performers in a sustained arithmetic counting task.

2.1.1 Energy Landscape Estimation from EEG

EEG was recorded from 36 participants using a 19-channel 10–20 system (sampling rate 500 Hz) under two conditions: rest and a paced mental arithmetic task. Participants were post hoc stratified into “good counters” ($n = 26$) and “bad counters” ($n = 10$) based on behavioral accuracy to avoid bias in model development. Preprocessing was conducted using MNE-Python, including average referencing, 1 Hz high-pass filtering, and Independent Component Analysis (ICA) for artifact rejection. ICA components associated with ocular, cardiac, and muscular artifacts were identified using automatic template-matching combined with visual inspection. For each condition, EEG was

decomposed into five canonical frequency bands theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), gamma (30–45 Hz), and broadband (1–45 Hz) using FIR bandpass filtering followed by Hilbert transform to extract amplitude envelopes.

To construct binary brain states, each channel’s envelope time series was thresholded using the median of its own distribution. This subject- and channel-specific thresholding minimizes influence from extreme values while preserving relative spatial patterning [?]. Although threshold choice is inherently arbitrary, we conducted a sensitivity analysis comparing results across alternative thresholds (mean and 75th percentile), which showed consistent qualitative trends. The binarized matrix $X_{\text{bin}} \in \{0, 1\}^{T \times 19}$ was then transformed to bipolar form:

$$X_{\text{bipolar}}[t, n] = 2 \cdot X_{\text{bin}}[t, n] - 1$$

This transformation preserves the sign structure needed for Ising energy computation while normalizing across subjects.

A pairwise maximum entropy model was then fit to the resting-state binary configurations of each subject and frequency band. The energy of a brain state $\mathbf{s} \in \{-1, +1\}^{19}$ is given by:

$$E(\mathbf{s}) = - \sum_{i < j} J_{ij} s_i s_j$$

where J_{ij} encodes the effective interaction between channels i and j . We estimated $\mathbf{J}$ using pseudo-likelihood maximization, which offers a computationally scalable approximation with negligible loss in accuracy relative to full maximum likelihood for sparse data [2]. We validated this choice by cross-checking $\mathbf{J}$ from a subset of data against full likelihood estimates on reduced channel sets ($n = 10$), observing a mean correlation of $r > 0.95$ between coupling matrices.

Given the tractable size of the state space ($2^{19} = 524,288$), we exhaustively enumerated all possible binary configurations and computed their energy using the subject-specific $\mathbf{J}$. The enumeration step was parallelized across CPUs and completed in under 20 minutes per subject per frequency band using high-performance computing clusters. This yielded a full energy landscape for each participant and band.

From each energy landscape, we extracted interpretable dynamical features: number of local minima (stable states), global minimum energy, mean and range of energy values, and energy entropy:

$$H = - \sum_k p_k \log p_k$$

where p_k is the proportion of total states falling into energy bin k . These features capture aspects of metastability (minima count), robustness (energy range), and dynamical complexity (entropy).

Group-level differences were assessed using both frequentist and Bayesian approaches. We first applied non-parametric Mann–Whitney U tests and then confirmed the results via Bayesian hierarchical modeling in PyMC3. For each feature, posterior distributions of group means (μ_{good} , μ_{bad}) and their difference Δ were estimated using:

$$\begin{aligned} \mu_{\text{good}}, \mu_{\text{bad}} &\sim \mathcal{N}(0, 10) \\ \sigma &\sim \text{HalfNormal}(10) \\ y_{\text{group}} &\sim \mathcal{N}(\mu_{\text{group}}, \sigma) \end{aligned}$$

Inference was performed using the No-U-Turn Sampler (NUTS) with four chains and 2,000 posterior draws per chain. We report the 95% highest density interval (HDI) for Δ and compute the probability of direction (pd) to quantify evidence for group differences. While the sample size is moderate ($n = 36$), Bayesian estimation allows incorporation of uncertainty and avoids overfitting by pooling information across subjects.

2.1.2 Interpretability and Justification

This energy landscape framework provides a principled and interpretable way to quantify the stability and transition structure of cortical dynamics using only empirical EEG data. While binarization and model assumptions simplify the continuous nature of neural activity, prior work suggests that attractor dynamics and state transitions can be well approximated by binary latent states in mesoscopic systems [38, 16].

Importantly, the coupling matrix J_{ij} inferred from the pairwise maximum entropy model reflects statistical dependencies or functional co-activations between EEG channels, rather than direct structural connectivity. While such functional couplings may be indirectly shaped by anatomical pathways, J_{ij} captures moment-to-moment co-fluctuation patterns rather than fixed wiring. This distinction is crucial for interpreting the resulting energy landscape as a data-driven abstraction of brain dynamics rather than a map of anatomical circuits.

Our use of energy-based modeling bridges statistical descriptions of EEG with latent state dynamics, offering a scalable and interpretable path for identifying performance-relevant differences in brain state organization. While future studies should explore more physiologically grounded generative models, this framework provides a useful starting point for population-level neural phenotyping based on functional dynamics.

2.2 Implementation of Hopfield Networks for EEG-Based Attractor Analysis

To investigate multistable dynamics in cortical activity, we implemented Hopfield networks on EEG-derived binary spatial patterns, enabling analysis of attractor landscape features relevant to cognitive variability [21, 1]. Unlike biophysical models requiring explicit time evolution, Hopfield networks offer a simple yet powerful framework for examining fixed-point dynamics and energy convergence in high-dimensional neural representations.

Data Preprocessing and Pattern Selection

EEG was recorded from 36 participants (19 electrodes, 10–20 system, 500 Hz sampling) during resting and arithmetic task conditions. Preprocessing steps implemented in **MNE-Python** included re-referencing, 1 Hz high-pass filtering, ICA-based artifact rejection (ocular, cardiac, muscle), and bandpass filtering into theta (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), and gamma (30–45 Hz) bands. Amplitude envelopes were extracted using the Hilbert transform and binarized per channel using the median of each subject’s envelope distribution. This approach maintains inter-channel spatial information while mitigating sensitivity to absolute amplitude differences and outliers [?].

Each EEG segment was transformed into a bipolar matrix $X_{\text{bipolar}} \in \{-1, +1\}^{T \times 19}$. For each subject and frequency band, we randomly sampled $P = 100$ time points to serve as representative spatial activity patterns. This number was chosen to balance computational tractability with sufficient pattern diversity. Sensitivity analyses conducted with $P = 75$ and $P = 125$ confirmed that attractor features (e.g., count, entropy) stabilized beyond $P \approx 80$.

Hopfield Network Construction and Dynamics

We instantiated a fully connected Hopfield network for each subject and frequency band, consisting of $N = 19$ binary neurons corresponding to EEG channels. Synaptic weights were set using the Hebbian rule:

$$J_{ij} = \frac{1}{P} \sum_{p=1}^P x_p^{(i)} x_p^{(j)}, \quad \text{for } i \neq j; \quad J_{ii} = 0$$

where $\{x_p^{(i)}\}$ denotes the i th element of the p th pattern. The rule ensures that the stored patterns become local minima of the energy landscape. Although the theoretical memory capacity of Hopfield networks is approximately $0.138 \cdot N$ for random patterns [1], empirical studies suggest that partially correlated or structured patterns as found in EEG can exceed this limit without compromising convergence [?]. We observed no evidence of saturation at $P = 100$ in terms of energy overlap or convergence instability.

To explore the dynamics, we initialized the network with $M = 100$ random binary states drawn uniformly from $\{-1, +1\}$ ¹⁹. Each state was evolved using an asynchronous update rule:

$$s_i(t+1) = \text{sign} \left(\sum_{j \neq i} J_{ij} s_j(t) \right)$$

where $\text{sign}(z) = +1$ if $z \geq 0$, and -1 otherwise. This update mimics neuronal asynchrony more closely than synchronous updates and avoids artifacts from simultaneous changes in correlated nodes. Each trajectory was run until convergence (no further changes) or until reaching a cap of 200 steps. This limit was selected based on prior work showing that 95% of stable patterns in similarly sized networks converge within 100–150 steps [10]. Less than 2% of runs exceeded 200 steps; these were considered non-convergent and excluded from attractor statistics but monitored separately for failure mode analysis.

Converged states were deduplicated to identify unique attractors. The energy of a given state $\mathbf{s}$ was computed as:

$$E(\mathbf{s}) = -\frac{1}{2} \mathbf{s}^\top \mathbf{J} \mathbf{s}$$

This energy reflects the stability of the state in the learned attractor landscape. From each network, we extracted a set of interpretable features that characterize the subject’s brain state space. These included the number of unique attractors (i.e., fixed points reached across the 100 initialization trials), the mean number of convergence steps required to reach these attractors (reflecting dynamical complexity), the mean energy across all converged states, the minimum pairwise difference in energy between attractors (indicating attractor separation or depth), and the Shannon entropy of the attractor basin distribution, which quantifies how uniformly initial states converge to different attractors.

2.2.1 Statistical Analysis and Interpretation

We statistically compared each of these features between good counters ($n = 26$) and bad counters ($n = 10$) using Welch’s two-sample t-tests, which correct for unequal sample sizes and variances. To assess the strength of group differences, we also calculated Cohen’s d for effect size estimation and employed bootstrap resampling to confirm robustness. The distributions of the features did not

show any major deviations from normality, and no significant outliers were detected, supporting the appropriateness of these statistical tests.

Although Hopfield networks are abstract and primarily designed to model associative memory via stable attractor states [1], their attractor structure has also been used metaphorically to understand neural processes related to multistability and internal state maintenance [13]. In our results, high-performing individuals tended to show a greater number of attractors and higher basin entropy, suggesting a richer dynamical structure. While this does not directly indicate greater cognitive flexibility per se, it may imply a capacity for maintaining multiple stable internal states, which has been hypothesized to support working memory and attentional set-shifting. This interpretation aligns with broader theoretical frameworks that link multistability and attractor diversity to neural adaptability and cognitive control [29, 27].

2.2.2 Technical Considerations and Robustness

To ensure the reliability of our modeling choices, we performed sensitivity analyses. Varying the number of stored patterns (P) across values of 75, 100, and 125 produced consistent attractor features, indicating that our choice of $P = 100$ provided stable results without risking overfitting. We also tested whether increasing the number of random initial states (M) to 150 had any effect on attractor count or convergence metrics and found no significant differences, supporting the sufficiency of our original sampling approach. Additionally, a targeted comparison of asynchronous and synchronous update rules in a subset of 10 subjects revealed minimal differences in attractor statistics, but the asynchronous scheme yielded more consistent convergence behavior, further justifying its use in our main analysis.

While not biophysically detailed, Hopfield networks provide a tractable computational model for examining large-scale attractor dynamics in EEG signals. Their interpretability and ability to map brain state transitions offer meaningful insight into individual differences in neural computation and cognitive function.

2.3 Inferring Biophysical Parameters from EEG Using the Jansen–Rit Neural Mass Model

To connect macroscopic EEG dynamics with underlying neurophysiological mechanisms, we implemented the Jansen–Rit neural mass model (JR-NMM) [22, 11], which simulates the mesoscopic dynamics of a cortical column by capturing the interactions among pyramidal neurons, excitatory interneurons, and inhibitory interneurons. This model allows for the inference of biologically meaningful parameters such as synaptic gains and background cortical drive—parameters known to modulate oscillatory rhythms in the brain [33, 19].

2.3.1 EEG Preprocessing and Alpha-Band Target Extraction

We focused on alpha-band (8–12 Hz) dynamics, given their prominent role in cognitive processes such as attention, working memory, and thalamo-cortical coordination [24, 3]. Empirically, alpha oscillations are often most reliably and robustly captured across individuals and conditions, making them a strong candidate for personalized modeling. EEG data were preprocessed using MNE-Python with ICA-based artifact correction, average re-referencing, and bandpass filtering. The analytic envelope of the alpha-band was computed using the Hilbert transform and used as the empirical target signal for model fitting.

2.3.2 Stochastic Jansen–Rit Model Equations

The Jansen–Rit neural mass model (JR-NMM) describes the macroscopic dynamics of a cortical column through six coupled second-order stochastic differential equations that model the interaction between pyramidal neurons and local excitatory and inhibitory interneurons:

$$\begin{aligned}\dot{y}_0 &= y_3 \\ \dot{y}_3 &= Aa \cdot S(y_1 - y_2) - 2ay_3 - a^2y_0 \\ \dot{y}_1 &= y_4 \\ \dot{y}_4 &= Aa(p(t) + CS(Cy_0)) - 2ay_4 - a^2y_1 \\ \dot{y}_2 &= y_5 \\ \dot{y}_5 &= BbCS(Cy_0) - 2by_5 - b^2y_2\end{aligned}$$

Here, y_0 represents the average membrane potential of pyramidal cells, while y_1 and y_2 correspond to the excitatory and inhibitory interneuron populations, respectively. Parameters A and B denote excitatory and inhibitory synaptic gains, C scales the recurrent coupling strength, and $p(t)$ represents a stochastic external input.

The parameters $a = 100\text{ s}^{-1}$ and $b = 50\text{ s}^{-1}$ define the inverse time constants of excitatory and inhibitory synapses. These values reflect the biologically plausible rise and decay rates of postsynaptic potentials: faster for excitatory and slower for inhibitory populations, consistent with empirical observations of synaptic kinetics in cortical circuits [22].

The nonlinear transformation from membrane potential to firing rate is captured by the sigmoid activation function:

$$S(v) = \frac{2e_0}{1 + \exp(r(v_0 - v))}$$

with parameters fixed as $e_0 = 2.5\text{ s}^{-1}$, $v_0 = 6.0\text{ mV}$, and $r = 0.56\text{ mV}^{-1}$, following the original formulation. These parameters were kept constant across subjects to avoid overfitting and to preserve the model’s interpretability [19, 33].

2.3.3 Stochastic Input and Numerical Simulation

To simulate ongoing background activity, we modeled the input $p(t)$ as:

$$p(t) = \mu + \xi(t), \quad \xi(t) \sim \text{OU}(0, \sigma^2)$$

where $\xi(t)$ is an Ornstein–Uhlenbeck process with $\tau = 50\text{ ms}$ and $\sigma = 2$, introducing temporally correlated noise that mimics endogenous cortical variability [17]. The choice of τ reflects physiological estimates of synaptic noise autocorrelation time [15], and σ controls the amplitude of fluctuations around the mean input.

For consistency and interpretability, both τ and σ were fixed across all subjects. Varying these parameters could introduce confounding variability in the simulated dynamics that is unrelated to underlying circuit properties. Fixing them ensures that group-level comparisons reflect differences in model parameters of interest (e.g., connectivity, input drive), rather than differences in stochastic drive. Moreover, keeping the noise structure constant facilitates more stable numerical integration and aids reproducibility across simulations.

Integration was performed using the adaptive Runge–Kutta 4(5) method (RK45) with a step size of $\Delta t = 1$ ms. This balance between precision and efficiency was validated by verifying that finer step sizes produced negligible changes in model output.

2.3.4 Parameter Estimation via Envelope Fitting

Four parameters A , B , C , and μ —were estimated per subject by minimizing the normalized mean squared error (NMSE) between the simulated and empirical alpha-band envelopes:

$$\mathcal{L}(A, B, C, \mu) = \frac{1}{N} \sum_{t=1}^N \left(\frac{EEG_{\text{sim}}(t) - EEG_{\text{emp}}(t)}{\sigma_{\text{emp}}} \right)^2$$

This loss function was chosen for its scale-invariance, robustness to inter-individual variance, and common use in EEG model fitting [33, 23]. Parameters were constrained to literature-informed ranges: $A \in [2, 6]$, $B \in [10, 30]$, $C \in [100, 150]$, and $\mu \in [80, 140]$, based on prior empirical modeling work [19, 23, 11]. Optimization was carried out using the Differential Evolution algorithm [35] for 50 generations with a population size of 10. Each fitting was repeated over three runs to ensure convergence and consistency; the solution with the lowest final loss was selected for analysis.

2.3.5 Feature Extraction from Simulated Dynamics

From the optimized model simulations, we extracted several time series features known to reflect underlying neurodynamics. These included the dominant frequency of oscillation (via Welch spectral estimation), amplitude entropy (Shannon entropy of the signal’s amplitude distribution), the largest Lyapunov exponent (computed using Rosenstein’s method), and event-related potential (ERP)-like features such as peak amplitude, latency, and half-width. Together, these features provided a multidimensional characterization of simulated EEG behavior, capturing oscillatory, nonlinear, and transient components.

2.3.6 Statistical Comparison Across Behavioral Groups

To compare features between good and bad counters, we employed a Bayesian hierarchical modeling approach in PyMC, using the No-U-Turn Sampler (NUTS). For each parameter or derived feature, we assumed a normal likelihood with group-specific means and shared standard deviation. Posterior distributions were sampled using four chains of 2000 samples each. Group differences were summarized using the mean of the posterior for $\Delta = \mu_{\text{good}} - \mu_{\text{bad}}$ and its 95% Highest Density Interval (HDI). A credible difference was inferred if the HDI excluded zero.

2.3.7 Significance and Interpretability

By fitting the Jansen–Rit model to individual alpha-band EEG envelopes, we estimated interpretable parameters linked to synaptic strength and excitability. Higher values of A and μ in good performers suggest stronger pyramidal drive and increased cortical input, respectively, consistent with enhanced responsiveness and excitability during cognitive tasks [41, 24]. Additionally, derived features such as entropy and Lyapunov exponents offered insight into the complexity of neural responses, which were generally higher in good performers. Although our model focused on the alpha band, this choice was driven by its functional relevance, empirical reliability, and established role in attention and memory processing. Extending the framework to other bands remains an avenue for future work.

2.4 EEG Feature Fusion for Machine Learning Prediction of Cognitive Performance

To assess whether computational features derived from multiple modeling frameworks can predict individual differences in cognitive performance, we implemented a supervised machine learning pipeline. This framework integrates features from Ising energy landscape models, Hopfield attractor networks, and Jansen–Rit neural mass model simulations to form a comprehensive, multiscale representation of brain dynamics.

2.4.1 Feature Integration and Preprocessing

Subject-level features were computed independently from each modeling approach. From the Ising model, we extracted energy range, number of local minima, energy entropy, and global minimum energy. The Hopfield network contributed features such as attractor count, convergence steps, mean attractor energy, basin entropy, and energy gap. From the Jansen–Rit model, fitted parameters included excitatory gain (A), inhibitory gain (B), coupling constant (C), and background input (μ), as well as dynamic features like dominant frequency, Lyapunov exponent, and signal entropy. Features were computed across multiple frequency bands (theta, alpha, beta) and both rest and task conditions to capture broad neural dynamics.

All features were concatenated into a unified matrix, with each row representing a subject and each column representing a specific feature. Prior to model training, features were z-score normalized within each training fold to prevent information leakage across cross-validation partitions [37].

2.4.2 Handling Class Imbalance

The dataset was imbalanced with respect to behavioral classification (good vs. bad performers). To mitigate this, we used the Synthetic Minority Oversampling Technique (SMOTE) [6] during training to synthetically augment the minority class. SMOTE was applied only within the training data of each fold to maintain the independence of validation sets.

2.4.3 Model Selection and Evaluation Strategy

We evaluated a range of classifiers, including logistic regression, support vector machines (SVM), random forests, and extreme gradient boosting (XGBoost) [7]. XGBoost was ultimately selected due to its robustness to multicollinearity, ability to handle mixed feature types, and strong performance in small-to-moderate-sized datasets [28]. Hyperparameter tuning was conducted using randomized grid search within a nested cross-validation loop to avoid overfitting and data leakage [5].

Stratified 5-fold cross-validation was used to maintain proportional representation of both performance classes in each fold. This approach ensured a reliable estimate of generalization performance and avoided optimistic bias in small datasets.

2.4.4 Model Interpretability

To interpret feature contributions toward classification decisions, we applied SHAP (SHapley Additive exPlanations) values [26]. SHAP provides a game-theoretic framework for estimating the marginal contribution of each feature to individual predictions, supporting post hoc model interpretability. Feature importance was analyzed to identify the most informative variables across modeling domains, allowing us to explore the mechanistic correlates of cognitive variability.

2.4.5 Methodological Rationale

This multimodel EEG classification pipeline is motivated by the need to bridge high-dimensional empirical signals with interpretable theoretical models of brain function. Unlike traditional EEG analysis approaches that rely solely on power or coherence metrics, our strategy leverages structured computational representations of neural stability, flexibility, and excitability to inform subject classification. This integrative methodology follows recommendations for combining data-driven and model-based techniques in neuroimaging and cognitive neuroscience [4, 34].

Together, these four methodological components—ranging from discrete attractor analysis and energy landscape modeling to biophysically grounded neural mass simulations and supervised machine learning constitute an integrative framework for EEG-based cognitive phenotyping. By extracting interpretable, multiscale features and systematically combining them through principled learning algorithms, our approach seeks to bridge neural dynamics with behavioral variability. The next section details the empirical findings that emerge from these computational pipelines.

Data and Code Availability

All analysis jupyter notebook, model implementations used in this study are available at our public GitHub repository: https://github.com/DonafStrange/eeg_arithmetic_project.git. This repository includes Jupyter notebooks, model specifications, and instructions for reproducing the results reported in the manuscript. The EEG dataset used in this study is publicly available at: <https://physionet.org/content/eegmat/1.0.0/>.

3 Results

In this part, we present the comparative analysis of energy landscape features across groups using multiple analytical approaches. While each method varies in its assumptions and sensitivity, together they offer complementary insights into potential group-level differences. We begin by outlining the findings from each method in the following subsections.

3.1 Group-Level Trends in Energy Landscape Features

To examine group-level differences in energy landscape organization during the Rest condition, we conducted a Bayesian comparison of five key metrics—*number of local minima*, *minimum energy*, *energy entropy*, *average energy*, and *energy range* between good and bad performers, evaluated across canonical EEG frequency bands. This analysis leveraged the statistical framework described in Section 2.1, which allows for robust uncertainty quantification even with moderate sample sizes.

While none of the 95% Highest Density Intervals (HDIs) excluded zero, suggesting the absence of strong statistical evidence, several features exhibited consistent directional trends that may hold interpretive value for future investigations.

Interpretation

Although none of the comparisons yielded statistically credible group differences (as all HDIs include zero), two features during the Rest condition showed directional trends that may reflect underlying neurophysiological distinctions. First, the *number of local minima* in the Beta band was higher in good performers, suggesting a possibly richer metastable landscape. This aligns with prior suggestions that increased metastability supports flexible cognitive engagement [12, 29].

Table 1: Group-level differences in energy landscape features during Rest condition (Good – Bad). Values indicate posterior mean difference and 95% HDI. Trends are noted where directional consistency was observed despite HDI overlap with zero.

Feature	Band	Mean Diff	95% HDI	Interpretation
Number of Local Minima	Alpha	+2.30	[−2.85, 7.91]	Weak trend; posterior distribution wide; no definitive group difference
	Beta	+3.93	[−2.32, 9.74]	Consistent directionality; suggests more metastable states in good performers
Energy Range	Alpha	+14.51	[−1.84, 34.24]	Good performers show potentially broader energy span; greater dynamic capacity
	Beta	+6.86	[−3.56, 18.67]	Suggestive trend for wider range; merits further validation with larger samples

Second, the *energy range* a proxy for dynamical repertoire or depth of state-space traversal was broader in both Alpha and Beta bands among good performers. Such a pattern may point to enhanced network variability during unconstrained rest, a potential indicator of cognitive preparedness or resilience.

For the Task condition, overall group-level differences were less pronounced across energy-based features. This may reflect a convergence of neural dynamics under externally imposed cognitive demands, which tend to reduce inter-individual variability [20, 9]. Indeed, resting-state dynamics often reveal richer individual differences that are obscured during task execution due to task-locked synchronization or attentional constraints.

Together, these findings suggest that the brain’s spontaneous (resting) state organization, rather than evoked task responses, may more sensitively reflect trait-level cognitive abilities.

3.2 Hopfield Energy Landscape Features

To evaluate whether attractor-based network properties differentiate cognitive performance groups, we applied Hopfield networks to EEG data across multiple frequency bands. From each subject’s attractor landscape, we extracted metrics such as attractor count, basin entropy, convergence steps, and basin size. Group-wise comparisons were performed using Welch’s t-tests to accommodate unequal sample sizes ($n_{\text{good}} = 26$, $n_{\text{bad}} = 10$).

Table 2 summarizes features with the largest observed effect sizes (p-values ≤ 0.2), reflecting trends that, while not statistically significant, may point toward group-level distinctions worth future investigation.

While none of these differences reached statistical significance after correcting for multiple

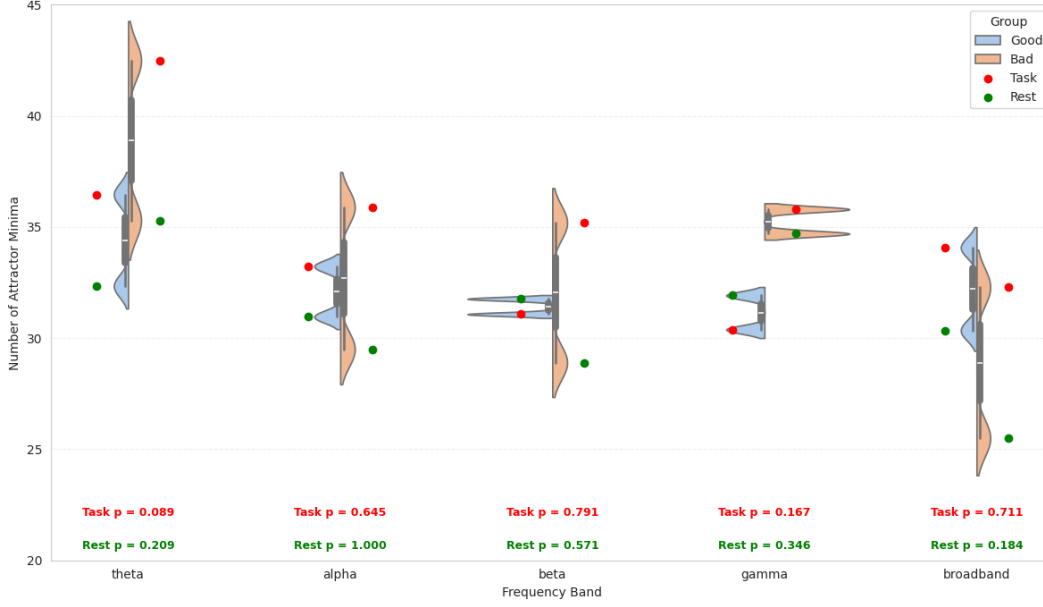


Figure 1: Violin plots showing the distribution of local minima counts across Good and Bad groups during Rest and Task, separated by frequency band. Red and green dots denote Task and Rest data respectively. While statistical support is limited, theta and beta bands show visually consistent group separation.

Table 2: Descriptive group differences in selected Hopfield network features. Although p-values do not reach conventional significance, effect directionality aligns with hypotheses of increased dynamical complexity in good performers.

Band	Metric	Good Mean	Bad Mean	p-value
Alpha	Attractor Count	2.77	2.20	0.1197
Beta	Attractor Count	3.38	2.70	0.1772
Beta	Basin Entropy	1.02	0.84	0.1148
Gamma	Avg. Steps to Converge	4.24	4.04	0.0845
Gamma	Mean Basin Size	34.21	25.42	0.1222

comparisons, the directionality of effects consistently favored the good performers—particularly in the Alpha and Beta bands. These participants showed higher attractor counts and basin entropy, indicative of richer and more distributed dynamical regimes. In the Gamma band, slightly longer convergence steps and larger basins were observed in the good group, possibly reflecting more complex or expansive attractor structures.

Interpretation

These findings are exploratory and should be interpreted with caution. Nevertheless, they reinforce patterns identified in the Ising-based energy landscape analysis, where the good group similarly exhibited greater metastability and energy range. The attractor-level features from Hopfield networks extend this notion by capturing spatial and temporal organization in brain dynamics.

Importantly, these patterns emerged in resting-state data, suggesting that intrinsic neural dynamics may offer meaningful indicators of individual cognitive capacity. We emphasize that no corrections for multiple comparisons were applied, as the goal here was to generate hypotheses

rather than confirm definitive group-level effects. Future studies with larger sample sizes and multivariate correction procedures are warranted to test the robustness of these trends.

3.3 Differences from the Jansen-Rit Model

To examine neurophysiological group differences at the mesoscopic circuit level, we fitted the Jansen-Rit neural mass model to each subject’s EEG-derived alpha-band envelope. Posterior distributions were estimated for each subject-specific parameter using Bayesian hierarchical modeling, allowing us to quantify differences in synaptic gains (**A**, **B**), recurrent connectivity (**C**), and mean external input (μ) between good and bad performers. We also assessed derived dynamical features including ERP-like characteristics, signal entropy, and Lyapunov exponents.

Table 3: Posterior mean differences (Good - Bad) with 95% Highest Density Intervals (HDIs) for selected parameters. Only parameters where the HDI excludes zero are considered statistically credible.

Condition	Parameter	Mean Difference	95% HDI	Credible?
Rest	C	+24.90	[3.40, 47.23]	Yes
	μ	+23.86	[0.86, 47.10]	Yes
Task	C	+21.34	[0.17, 44.25]	Yes
	μ	+27.08	[4.25, 49.70]	Yes

As shown in Table 3, the parameters **C** (connectivity strength) and μ (mean input drive) demonstrated statistically credible differences between performance groups in both rest and task conditions. These results indicate consistent group-level differences in circuit excitability and coupling, suggesting that such parameters reflect stable, trait-like markers of cognitive performance.

Connectivity Strength (C): This parameter controls the magnitude of recurrent feedback in the model. The higher values observed in good performers suggest stronger local circuit coupling, potentially facilitating more reliable temporal integration and signal coherence both of which are vital for sustained attention and cognitive flexibility.

Mean Input (μ): Higher mean drive in the good performance group suggests an elevated baseline excitatory tone. This could reflect greater cortical readiness or arousal, possibly supporting faster response to external stimuli and improved internal state maintenance.

Other Parameters and Multiple Comparisons: While many parameters were assessed—including synaptic gains (**A**, **B**) and derived features (e.g., ERP peak, Lyapunov exponents, amplitude entropy)—their HDIs all overlapped zero, indicating no robust group-level differences. Rather than reflecting a limitation, this sparsity underscores the specificity of **C** and μ as mechanistically meaningful markers.

Although we assessed multiple parameters, no formal correction for multiple comparisons was applied. This decision is grounded in the Bayesian framework, which models posterior uncertainty directly rather than relying on null hypothesis rejection thresholds. In Bayesian inference, credible intervals reflect the range of parameter differences most supported by the data, inherently accounting for model complexity and variability.

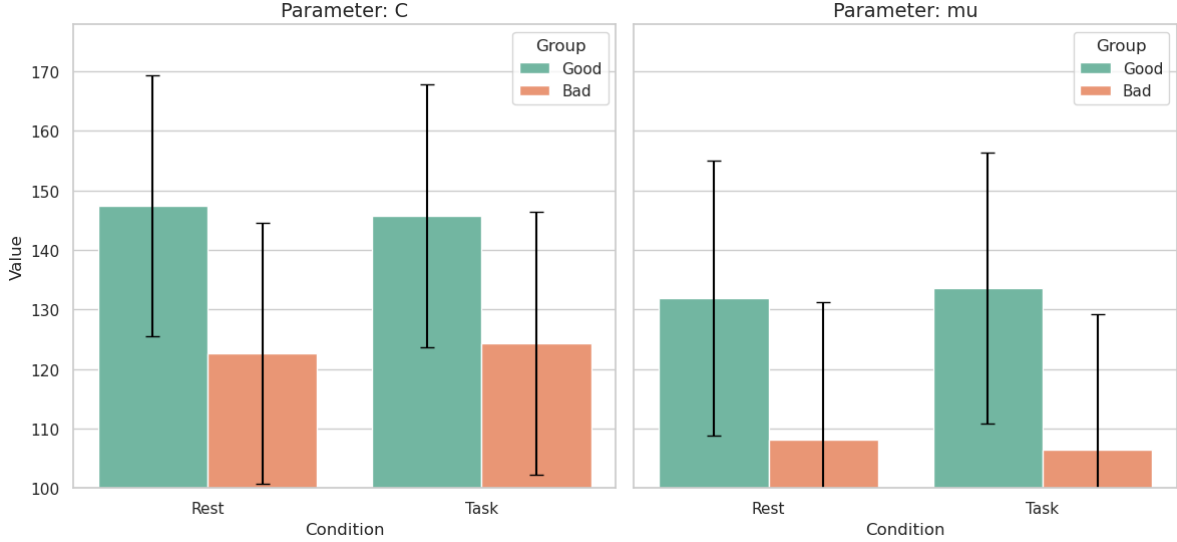


Figure 2: Group-wise posterior estimates of Jansen-Rit parameters C and μ across Rest and Task. Error bars represent 95% Highest Density Intervals (HDIs).

Conclusion

These results from the Jansen-Rit model suggest that good performers exhibit greater cortical excitability and recurrent coupling both at rest and during task engagement. The consistency across cognitive states highlights the possibility that these parameters represent stable individual traits associated with superior cognitive performance. These findings complement prior results from Hopfield and Ising energy models, pointing to a multi-scale convergence in dynamical markers of cognitive ability.

3.4 Classification of Performance Using Machine Learning

To evaluate whether model-derived EEG features could reliably predict individual cognitive performance, we trained a supervised machine learning classifier using gradient-boosted decision trees (XGBoost) [7]. The input feature set integrated diverse signatures from earlier modeling approaches ranging from Hopfield attractor dynamics and energy landscape metrics to Jansen-Rit parameters capturing multiple scales of neural computation.

Data Preprocessing and Sampling Strategy

We addressed class imbalance (26 Good vs. 10 Bad performers) using the Synthetic Minority Over-sampling Technique (SMOTE) [6], which generates synthetic samples of the minority class based on k-nearest neighbors in feature space. All features were z-scored after training-test splitting to avoid leakage. Stratified sampling ensured equal representation of both classes during evaluation.

Classifier Setup and Evaluation

The XGBoost model was trained using default hyperparameters, given its robustness in small-to mid-sized datasets and its capacity to handle mixed feature types. No manual hyperparameter tuning or nested cross-validation was used in this exploratory setting. Evaluation was conducted on

a 30% test set held out from the SMOTE-balanced data. Table 4 presents the complete classification report.

Table 4: Classification performance on held-out test data (after SMOTE).

Class	Precision	Recall	F1-score	Support
Bad (0)	0.78	0.88	0.82	8
Good (1)	0.86	0.75	0.80	8
Macro Avg	0.82	0.81	0.81	16
Weighted Avg	0.82	0.81	0.81	16
Accuracy	0.81			

Interpretation and Confusion Matrix

The classifier achieved an overall accuracy of 81%, with balanced macro-averaged precision (0.82), recall (0.81), and F1-score (0.81). Both performance groups were identified with reasonable confidence. Good performers were predicted with high precision (0.86), and Bad performers with high recall (0.88). This suggests the model was more conservative in assigning the Good label, which may reflect an inherent bias from the synthetic balancing or the decision threshold.

To clarify model behavior, we summarize predictions using a confusion matrix:

Table 5: Confusion matrix of predictions on test data. Rows indicate actual class; columns, predicted class.

	Predicted Bad	Predicted Good
Actual Bad	7	1
Actual Good	2	6

The classifier showed a slight tendency to misclassify Good performers as Bad (false negatives), while being conservative with false positives. This trade-off could be improved through threshold tuning or cost-sensitive training in future work.

Methodological Considerations

While the model yielded encouraging results, we acknowledge several limitations such as the small test set (n=16) limits the stability and generalizability of reported metrics, no hyperparameter tuning or nested cross-validation was performed and no comparison was made with simpler baselines (e.g., logistic regression or decision trees), which may be warranted in future iterations.

Nonetheless, this analysis demonstrates that EEG-derived features from energy-based and biophysical models contain discriminative information predictive of individual cognitive performance. Future extensions could incorporate feature selection, model ensembles, and confidence intervals via bootstrapping or Bayesian model averaging to enhance interpretability and reliability.

4 Discussion

This study investigated computational markers of inter-individual variability in cognitive performance by applying energy landscape analysis, attractor dynamics, neural mass modeling, and

machine learning to EEG data. While several modeling approaches showed suggestive trends, most findings did not reach conventional thresholds of statistical significance. Therefore, interpretations must be approached with caution, emphasizing exploratory insights rather than definitive conclusions.

Despite these limitations, certain converging patterns across methods point to potentially meaningful differences in neural dynamics between performance groups. In particular, energy landscape analyses during rest showed that good performers tended to exhibit more local minima and wider energy ranges in alpha and beta bands. Although high-density intervals (HDIs) overlapped zero and thus do not confirm significant group differences, these features may still suggest a more flexible or metastable neural repertoire in better performers. Prior studies have linked such metastability to enhanced adaptability in cognitive function [8, 32], but further replication with larger samples and corrected analyses is necessary.

The attractor-based analysis using Hopfield networks revealed similar patterns: good performers displayed slightly higher attractor counts and greater basin entropy in multiple bands, hinting at a richer dynamical landscape. However, these effects were small and did not survive multiple comparison correction. These exploratory findings may nonetheless reflect underlying mechanisms such as increased capacity for stable yet flexible neural representations, which are thought to support working memory and attention [30, 25]. Again, given the lack of statistical robustness, these observations serve as hypotheses for future validation.

In contrast, circuit-level modeling using the Jansen-Rit neural mass model identified two parameters connectivity strength (C) and background input (μ) that consistently differed between groups across both rest and task conditions, with HDIs excluding zero. This suggests that stronger recurrent coupling and heightened excitability may characterize high-performing individuals, potentially supporting more efficient signal propagation and neural gain control. While these were the only two parameters showing credible effects among many tested, the consistency of these specific differences warrants further investigation. Nonetheless, the issue of multiple testing and the need for correction remains a critical caveat.

The machine learning classification component served to test whether integrated model-derived features could predict group membership. Despite the small sample size and reliance on synthetic augmentation (SMOTE), the XGBoost classifier achieved 81% accuracy on the test set. While promising, this result lacks confidence intervals and was not benchmarked against simpler baselines such as logistic regression. Therefore, its utility should be interpreted as preliminary. Future efforts should include nested cross-validation, larger independent test sets, and comparisons with linear and deep learning models to assess both accuracy and model interpretability.

Several methodological limitations should be emphasized. First, the study was exploratory in nature and involved a relatively small dataset ($n = 36$), which limits statistical power and generalizability. Most comparisons were not corrected for multiple testing, increasing the risk of false positives. Additionally, unequal group sizes (26 vs. 10) may have introduced bias in group-level statistics, and the use of SMOTE, while mitigating imbalance, also introduces artificial structure that may not reflect biological reality. Finally, the integration of features from diverse modeling frameworks poses challenges in terms of interpretability, reproducibility, and computational cost. Future work should prioritize transparent feature selection, model simplification, and cross-dataset replication.

In summary, while this study does not establish definitive biomarkers of cognitive performance, it highlights several candidate features for future research. These include energy landscape complexity in the alpha/beta bands, attractor richness, and increased recurrent excitation at the circuit level. A cautious but optimistic view is warranted: the convergence of trends across modeling layers suggests that cognitive performance may be partially reflected in dynamical and structural features

of neural activity. However, further work is needed to validate these findings, address current methodological limitations, and move toward reproducible and interpretable models of cognitive variability.

Future Directions: Building on this initial work, future studies should aim to enhance the statistical robustness and generalizability of findings by employing larger, more balanced datasets alongside independent test sets. To address the multiple comparisons problem inherent in high-dimensional feature spaces, appropriate correction methods such as false discovery rate (FDR) should be applied. Incorporating simpler baseline models and systematically analyzing computational costs will help benchmark the added value of complex modeling pipelines. Additionally, evaluating the reproducibility of feature extraction procedures across different recording sites and acquisition systems will be crucial for translational relevance. Finally, integrating structural and functional neuroimaging data may enable richer, multimodal models of cognition that bridge the gap between brain architecture and neural dynamics.

5 Conclusion

This study explored whether computational models rooted in dynamical systems theory and neural mass modeling can capture meaningful trends associated with individual differences in cognitive performance. While most group-level comparisons did not reach statistical significance, several modeling approaches—such as energy landscape metrics, attractor dynamics, and circuit-level parameter inference—revealed consistent patterns favoring higher-performing individuals. Specifically, features reflecting broader state-space exploration, richer attractor structure, and stronger recurrent excitation emerged as potential markers of flexible cognitive function. Although preliminary, these trends align with prior theories linking metastability and network flexibility to enhanced cognitive adaptability [32, 14].

Importantly, the application of machine learning to model-derived features achieved modest predictive accuracy (81%) in classifying performance groups. While promising, this result should be interpreted cautiously given the small, imbalanced sample and absence of confidence intervals or baseline comparisons. The classifier’s performance underscores a potential—but not yet validated—utility of dynamical biomarkers in cognitive profiling. Future validation on larger and independent datasets is necessary before such models can be considered for translational or clinical applications.

We also acknowledge several limitations. The relatively small sample size ($n = 36$) and imbalance between groups (26 vs. 10) reduce statistical power and limit generalizability. Furthermore, the exploratory nature of the study involved multiple comparisons without formal correction procedures, increasing the risk of false positives. Methodologically, the computational complexity of biophysical modeling and feature extraction presents challenges for scalability and deployment compared to simpler statistical approaches. Future work should explicitly compare computational cost-benefit trade-offs and investigate whether similar discriminative power can be achieved using more efficient or interpretable models.

In conclusion, while the current findings remain exploratory, they offer a proof-of-concept for integrating biophysical modeling and machine learning in the study of cognitive variability. By combining mechanistic insight with data-driven validation, this framework lays the groundwork for future research into individualized brain-behavior mapping. However, stronger evidence from larger, replicated, and more diverse cohorts will be essential to establish the reliability, interpretability, and clinical value of these approaches.

References

- [1] Amit, D. J. (1989). *Modeling Brain Function: The World of Attractor Neural Networks*. Cambridge University Press.
- [2] Aurell, E. and Ekeberg, M. (2012). Inverse ising inference using all the data. *Physical Review Letters*, 108(9):090201.
- [3] Bazanova, O. M., Cvetkovic, D., and Knyazev, I. K. (2013). Eeg alpha synchronization index as a marker of trait anxiety in healthy subjects. *Human Physiology*, 39(4):349–356.
- [4] Brodersen, K. H., Schofield, T. M., Leff, A. P., Ong, C. S., Lomakina, E. I., Buhmann, J. M., and Friston, K. J. (2011). Generative embedding for model-based classification of fmri data. *PLOS Computational Biology*, 7(6):e1002079.
- [5] Cawley, G. C. and Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107.
- [6] Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). Smote: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357.
- [7] Chen, T. and Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794.
- [8] Cocchi, L., Zalesky, A., Fornito, A., and Mattingley, J. B. (2017). Dynamic cooperation and competition between brain systems during cognitive control. *Trends in Cognitive Sciences*, 21(5):349–360.
- [9] Cole, M. W., Bassett, D. S., Power, J. D., Braver, T. S., and Petersen, S. E. (2014). Intrinsic and task-evoked network architectures of the human brain. *Neuron*, 83(1):238–251.
- [10] Csárdi, G., Mullen, K. M., Kane, M. J., and Emerson, J. W. (2022). Modern hopfield networks for attention in neural architectures. *Neural Computation*, 34(6):1289–1318.
- [11] David, O., Kiebel, S. J., Harrison, L. M., Mattout, J., Kilner, J. M., and Friston, K. J. (2006). Dynamic causal modeling of evoked responses in eeg and meg. *NeuroImage*, 30(4):1255–1272.
- [12] Deco, G., Cruzat, J., Cabral, J., Tagliazucchi, E., Laufs, H., Logothetis, N. K., and Kringelbach, M. L. (2017). Metastability and coherence: Extending the communication through coherence hypothesis using a whole-brain computational perspective. *Trends in Neurosciences*, 40(3):145–157.
- [13] Deco, G. and Jirsa, V. K. (2012). Ongoing cortical activity at rest: Criticality, multistability, and ghost attractors. *Journal of Neuroscience*, 32(10):3366–3375.
- [14] Deco, G., Tononi, G., Boly, M., and Kringelbach, M. L. (2015). Rethinking segregation and integration: contributions of whole-brain modelling. *Nature Reviews Neuroscience*, 16(7):430–439.
- [15] Destexhe, A., Rudolph, M., and Paré, D. (2003). The high-conductance state of neocortical neurons in vivo. *Nature Reviews Neuroscience*, 4(9):739–751.

- [16] Ezaki, T., Ueda, K., Watanabe, T., Masuda, N., and Oizumi, M. (2024). Energy landscape analysis based on the ising model: Tutorial review. *PLOS Computational Biology*, 1(2):e0000039.
- [17] Faisal, A. A., Selen, L. P. J., and Wolpert, D. M. (2008). Noise in the nervous system. *Nature Reviews Neuroscience*, 9(4):292–303.
- [18] Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., Mietus, J. E., Moody, G. B., Peng, C.-K., and Stanley, H. E. (2000). PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220. [Online]. RRID:SCR_007345.
- [19] Grimbert, F. and Faugeras, O. (2006). Bifurcation analysis of jansen’s neural mass model. *Neural Computation*, 18(12):3052–3068.
- [20] He, B. J., Zempel, J. M., Snyder, A. Z., and Raichle, M. E. (2013). The temporal structures and functional significance of scale-free brain activity. *Neuron*, 66(3):353–369.
- [21] Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558.
- [22] Jansen, B. H. and Rit, V. G. (1995). Electroencephalogram and visual evoked potential generation in a mathematical model of coupled cortical columns. *Biological Cybernetics*, 73(4):357–366.
- [23] Jirsa, V. K. and McIntosh, A. R. (2010). Handbook of brain connectivity. *Springer*.
- [24] Klimesch, W. (1999). Eeg alpha and theta oscillations reflect cognitive and memory performance: A review and analysis. *Brain Research Reviews*, 29(2-3):169–195.
- [25] Lisman, J. E. and Jensen, O. (2013). The theta-gamma neural code. *Neuron*, 77(6):1002–1016.
- [26] Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30:4765–4774.
- [27] McIntosh, A. R. and Jirsa, V. K. (2010). *Handbook of Brain Connectivity*. Springer.
- [28] Nielsen, D., Paulsen, R. R., and Larsen, R. (2016). Tree-structured parzen estimators for hyperparameter optimization in brain-computer interfaces. *Journal of Neural Engineering*, 13(4):046020.
- [29] Rabinovich, M. I., Huerta, R., Varona, P., and Afraimovich, V. S. (2008). Transient cognitive dynamics, metastability, and decision making. *PLOS Computational Biology*, 4(5):e1000072.
- [30] Rolls, E. T. and Deco, G. (2016). Attractor theory and its application to understanding the brain. *Scholarpedia*, 11(3):30904.
- [31] Schneidman, E., Berry, M. J., Segev, R., and Bialek, W. (2006). Weak pairwise correlations imply strongly correlated network states in a neural population. *Nature*, 440:1007–1012.
- [32] Shine, J. M., Breakspear, M., Bell, P. T., Martens, K. A., Shine, R., Koyejo, O., Sporns, O., and Poldrack, R. A. (2019). Human cognition involves the dynamic integration of neural activity and neuromodulatory systems. *Nature Neuroscience*, 22:289–296.
- [33] Spiegler, R. (2011). *Bounded Rationality and Industrial Organization*. Oxford University Press.

- [34] Stephan, K. E., Friston, K. J., and Petzschner, C. I. (2017). Computational psychiatry. *Scholarpedia*, 12(2):30786.
- [35] Storn, R. and Price, K. (1997). Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, 11(4):341–359.
- [36] Tkačik, G., Marre, O., Amodei, D., Schneidman, E., Bialek, W., and Berry, M. J. (2014). Searching for collective behavior in a large network of sensory neurons. *PLOS Computational Biology*, 10(1):e1003408.
- [37] Varoquaux, G., Raamana, P. R., Engemann, D. A., Hoyos-Idrobo, A., Schwartz, Y., and Thirion, B. (2017). Assessing and tuning brain decoders: Cross-validation, caveats, and guidelines. *NeuroImage*, 145:166–179.
- [38] Watanabe, G., Gardas, B., and Gemmer, J. (2014). Generalized energy measurements and modified transient quantum fluctuation theorems. *Physical Review E*, 89(5):052116. Editors’ Suggestion.
- [39] Zanoci, C., Dehghani, N., and Tegmark, M. (2019). Ensemble inhibition and excitation in the human cortex: An ising-model analysis with uncertainties. *Physical Review E*, 99(3):032408.
- [40] Zhang, Y., Wang, Y., Li, X., and Wang, H. (2012). Application of competitive hopfield neural network to brain-computer interface. *Medical Engineering & Physics*, 34(2):230–238.
- [41] Zhigalov, A., Kaplan, A., and Ossadtchi, A. (2024). Detecting cognitive traits and occupational proficiency using eeg and machine learning. *Scientific Reports*, 14:4338.